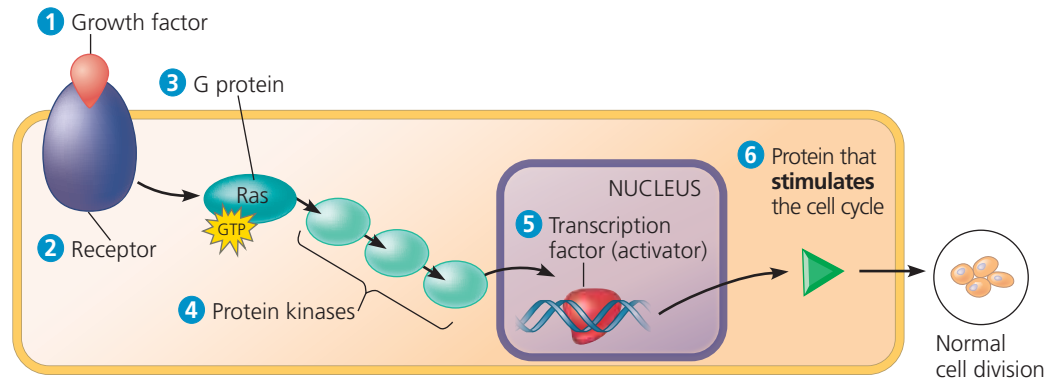


▼ **Figure 18.24 Normal and mutant cell cycle–stimulating pathway.**

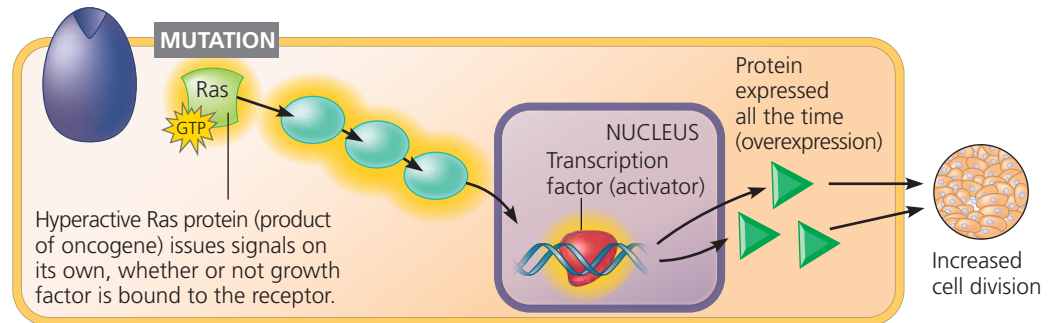
(a) Normal cell cycle–stimulating pathway.

The normal pathway is triggered by **1** a growth factor that binds to **2** its receptor in the plasma membrane. The signal is relayed to **3** a G protein called Ras. Like all G proteins, Ras is active when GTP is bound to it. Ras passes the signal to **4** a series of protein kinases. The last kinase activates **5** a transcription factor (activator) that turns on one or more genes for **6** a protein that stimulates the cell cycle.



(b) Mutant cell cycle–stimulating pathway.

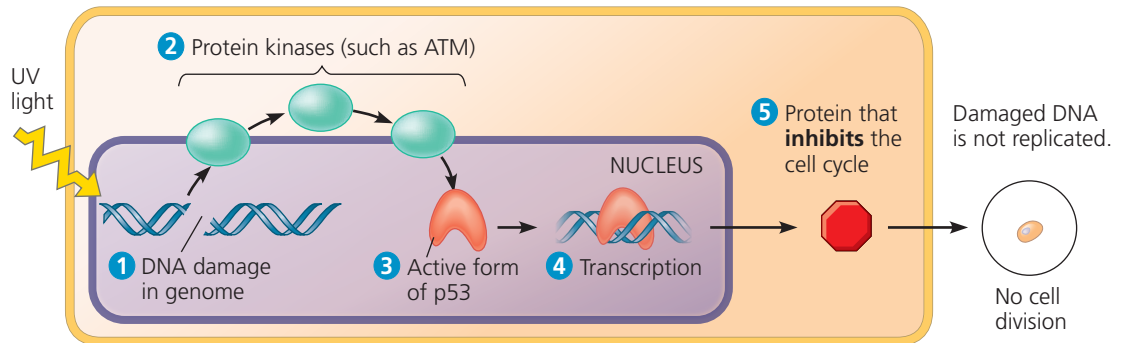
If a mutation makes Ras or any other pathway component abnormally active, excessive cell division and cancer may result.



▼ **Figure 18.25 Normal and mutant cell cycle–inhibiting pathway.**

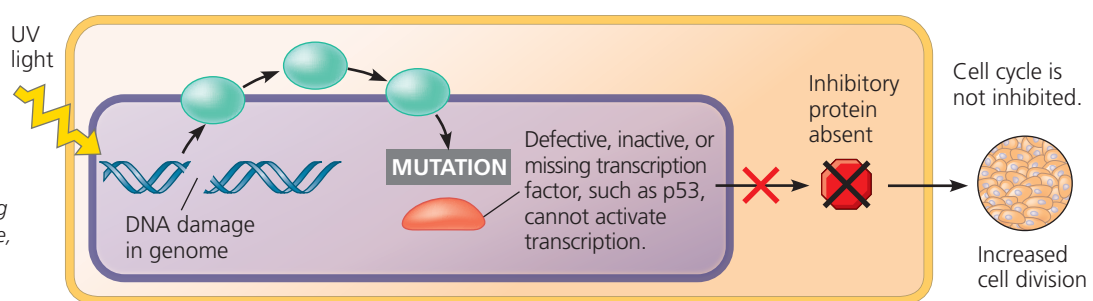
(a) Normal cell cycle–inhibiting pathway.

In the normal pathway, **1** DNA damage is an intracellular signal that is passed via **2** protein kinases, leading to activation of **3** p53. **4** Activated p53 promotes transcription of the gene for **5** a protein that inhibits the cell cycle. The resulting suppression of cell division ensures that the damaged DNA is not replicated. If the DNA damage is irreparable, then the p53 signal leads to apoptosis.



(b) Mutant cell cycle–inhibiting pathway.

Mutations causing deficiencies in any pathway component can contribute to the development of cancer.



? Explain whether a cancer-causing mutation in a tumor-suppressor gene, such as p53, is more likely to be a recessive or a dominant mutation.

called *ATM*) can lead to excessive cell growth and cancer (**Figure 18.25b**).

The *p53* gene has been called the “guardian angel of the genome.” Once the p53 protein is activated—say, by the ATM protein, a protein kinase, after DNA damage—p53 activates several other genes, such as *p21*. The p21 protein halts the cell cycle by binding to cyclin-dependent kinases, allowing time

for the cell to repair the DNA. Researchers recently showed that p53 also activates expression of a group of miRNAs that inhibit the cell cycle. The p53 protein can also turn on genes directly involved in DNA repair. If DNA damage is irreparable, p53 activates “suicide” genes, whose protein products bring about programmed cell death (apoptosis; see Figure 11.20). Thus, p53 acts in several ways to prevent a cell from passing